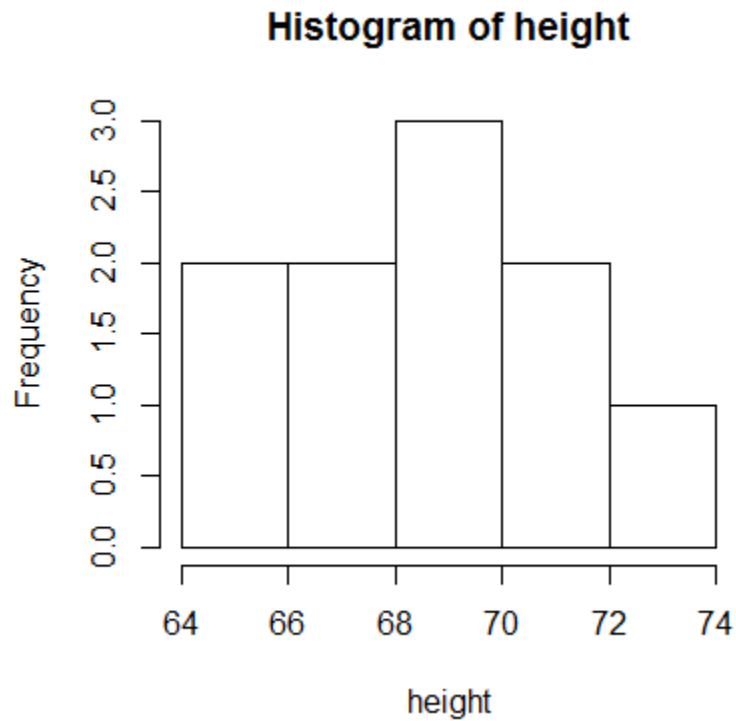


1.2

(a)



(b)

6 | 4

6 | 6789

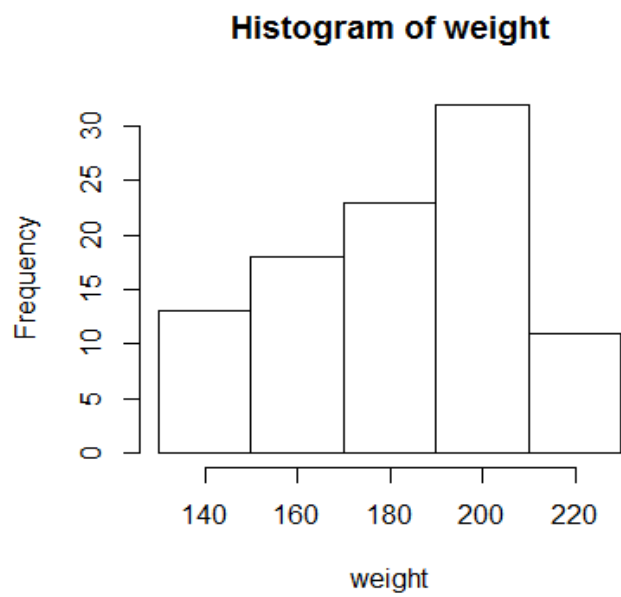
7 | 00114

1.4

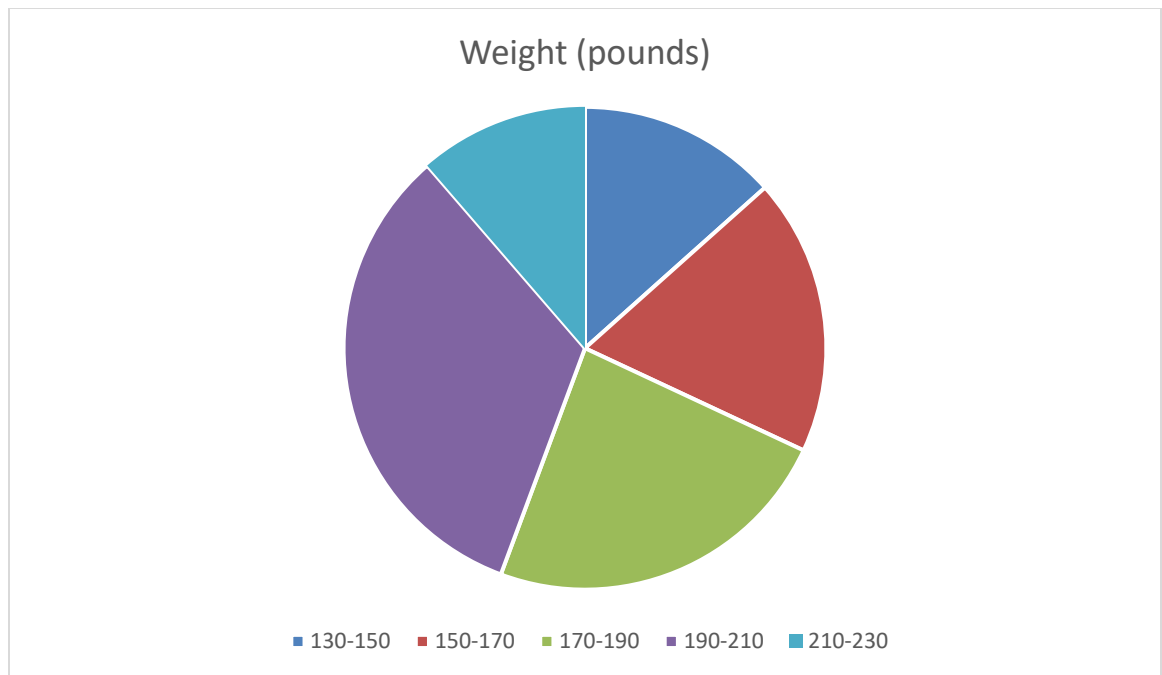
- (a) sample mean: 67.3, sample median: 57.5, sample standard deviation: 37.7
 Because it is skewed to the right, the mean should be larger than the median.
 Obviously 37.7 is too small to be either the mean or the median.
- (b) skewed to the right

1.5

(a)



(b)



(c) Since $n = 97$ is odd, the median is the middle value which is $x_{(n+1/2)} =$

$$x_{(98/2)} = x_{(49)}.$$

The 49th observation is in the class interval of 170-190.

1.7

(a)

$$\bar{x} = \frac{1}{7} \sum_{i=1}^7 x_i = \frac{9 + 8 + 12 + 5 + 3 + 6 + 13}{7} = 8$$

(b)

$$s^2 = \frac{\sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2/n}{n-1} = \frac{528 - 56^2/7}{6} = 13.3$$

$$s = \sqrt{13.3} = 3.65$$

1.10

(a)

$$\bar{x} = \frac{1}{12} \sum_{i=1}^{12} x_i = 62$$

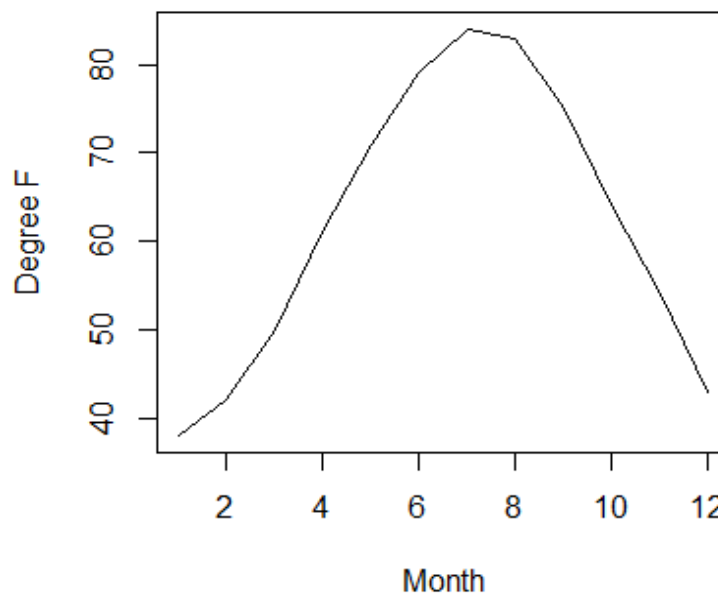
$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} = \frac{(38-62)^2 + (42-62)^2 + \dots + (43-62)^2}{11} = 274$$

(b)

$$s^2 = \frac{\sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2/n}{n-1} = \frac{49142 - 744^2/12}{11} = 274$$

(c)

High Temperature



(d)

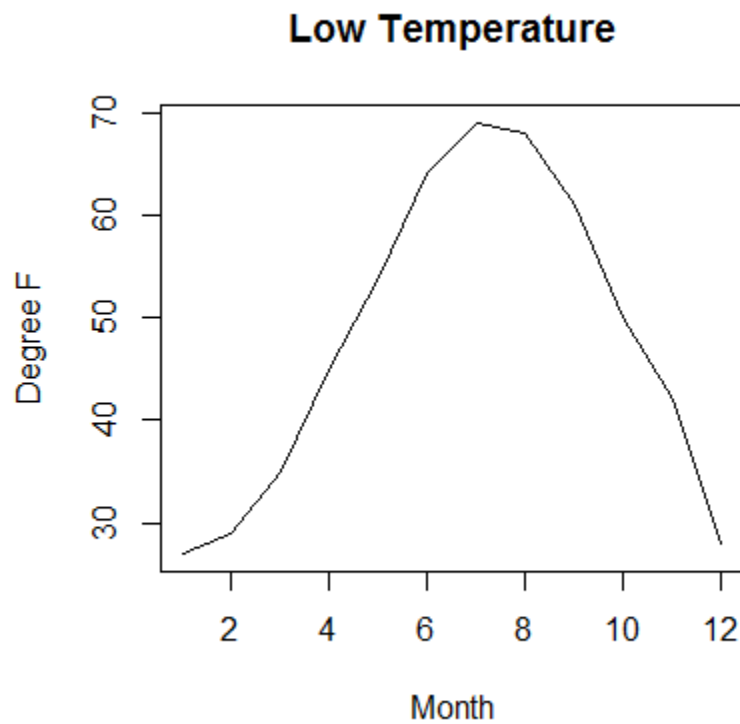
$$\bar{y} = \frac{1}{12} \sum_{i=1}^{12} y_i = 47.67$$

$$s^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n - 1} = \frac{(27 - 47.67)^2 + (29 - 47.67)^2 + \dots + (28 - 47.67)^2}{11} = 247.33$$

(e)

$$s^2 = \frac{\sum_{i=1}^n y_i^2 - (\sum_{i=1}^n y_i)^2/n}{n - 1} = \frac{29986 - 572^2/12}{11} = 247.33$$

(f)



1.13

- (a) Since $n = 50$ is even, the median is the average of the middle two values:

$$\frac{x_{(n/2)} + x_{(\frac{n}{2}+1)}}{2} = \frac{x_{(50/2)} + x_{(\frac{50}{2}+1)}}{2} = \frac{x_{(25)} + x_{(26)}}{2} = \frac{81 + 81}{2} = 81$$

- (b) First quartile: $np = 50(0.25) = 12.5 \uparrow 13$ $Q_1 = x_{(13)} = 77$

Third quartile: $np = 50(0.75) = 37.5 \uparrow 38$ $Q_3 = x_{(38)} = 85$

- (c)

$$\bar{x} = \frac{1}{50} \sum_{i=1}^{50} x_i = \frac{4066}{50} = 81.32$$

- (d)

$$s^2 = \frac{\sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2/n}{n-1} = 46.2$$

$$s = \sqrt{46.2} = 6.8$$

- (e) $(\bar{x} - 2s, \bar{x} + 2s) = (81.32 - 2(6.8), 81.32 + 2(6.8)) = (67.72, 94.92)$

$\frac{47}{50} = 0.94$ of the scores lie in the interval.

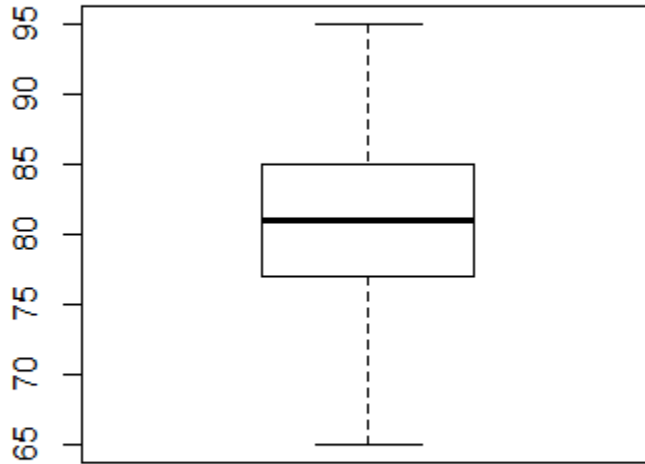
- (f) Sample range = $95 - 65 = 30$

Interquartile range = $Q_3 - Q_1 = 85 - 77 = 8$

- (g) $np = 50(0.9) = 45$ is an integer. The 90th percentile is

$$\frac{x_{(45)} + x_{(46)}}{2} = \frac{91 + 93}{2} = 92.$$

(h)



(i) $Q_1 - 1.5 \text{ IQR} = 77 - 1.5(8) = 77 - 12 = 65$

$Q_3 + 1.5 \text{ IQR} = 85 + 12 = 97$

No score is below 65 or above 97. Thus, there are no outliers.

1.18

(a) The values are sorted as follows:

60 60 61 61 61 62 63 63 64 64 64 64 65 65 65 66 66 66 66 66 66 67 67 67 67
67 67 67 67 67 67 67 67 68 68 68 68 68 68 69 69 69 69 69 70 70 70 71 72 72

Since $n = 50$ is even, the median is the average of the middle two values:

$$\frac{x_{(n/2)} + x_{(\frac{n}{2}+1)}}{2} = \frac{x_{(50/2)} + x_{(\frac{50}{2}+1)}}{2} = \frac{x_{(25)} + x_{(26)}}{2} = \frac{67 + 67}{2} = 67$$

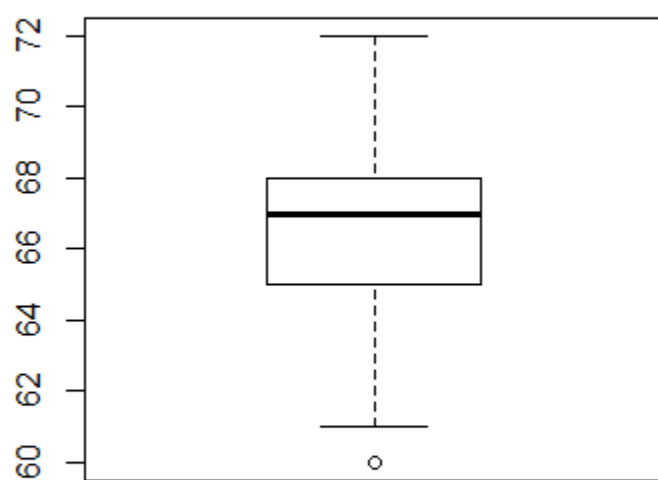
First quartile: $np = 50(0.25) = 12.5 \uparrow 13$ $Q_1 = x_{(13)} = 65$

Third quartile: $np = 50(0.75) = 37.5 \uparrow 38$ $Q_3 = x_{(38)} = 68$

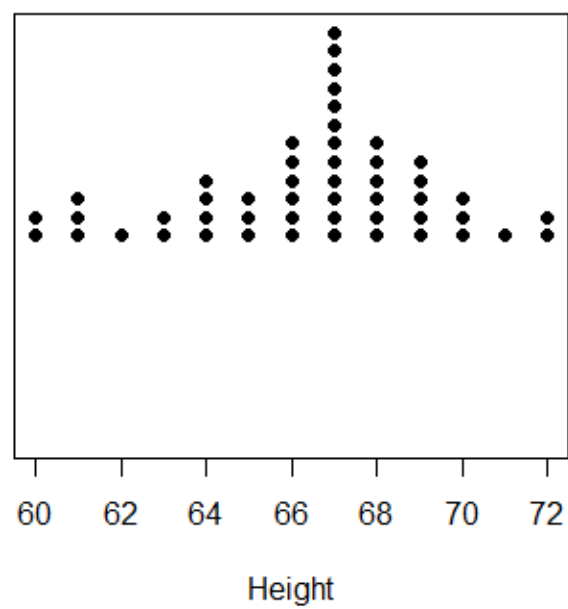
(b) Sample range = $72 - 60 = 12$

Interquartile range = $Q_3 - Q_1 = 68 - 65 = 3$

(c)



(d)



(e) $Q_1 - 1.5 \text{ IQR} = 65 - 1.5(3) = 65 - 4.5 = 60.5$

$Q_3 + 1.5 \text{ IQR} = 68 + 4.5 = 72.5$

All but two observations with value 60 are included in (60.5, 72.5). Thus, 60 is an outlier.

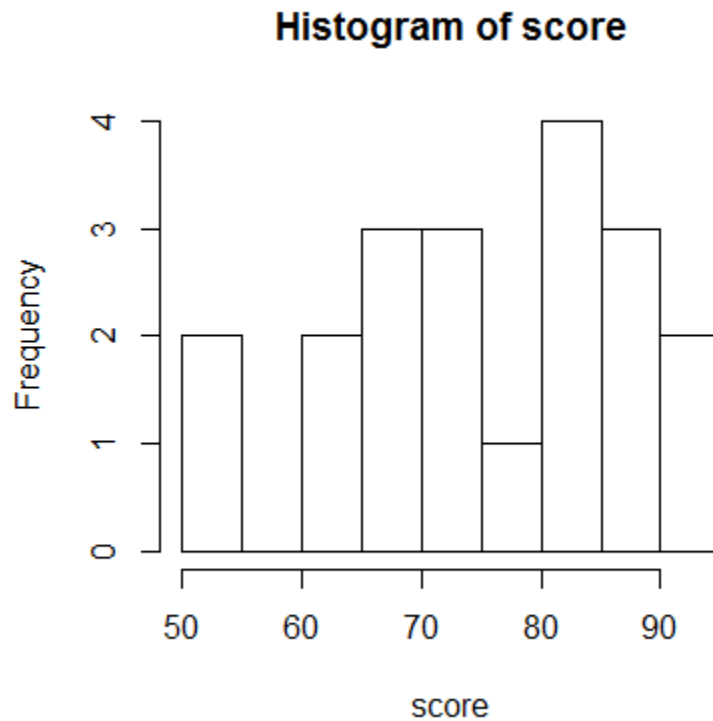
1.22

- (a) Both distributions are skewed to the left. Overall, the second midterm scores are higher than the first midterm scores. However, the outliers in the second boxplot show that several students fell behind.
- (b) Yes.
- (c) The second midterm has a larger difference between the mean and the median because it is more skewed. The mean is lower than the median because it is skewed to the left.
- (d) Midterm 2 has a larger standard deviation of the scores.

1.23

(a)

```
>score=c(69,84,52,93,81,74,89,85,88,63,87,64,67,72,74,55,82,91,68,77)
>hist(score)
```



```

(b) >stem(score)
  5 | 25
  6 | 34789
  7 | 2447
  8 | 1245789
  9 | 13
(c) >mean(score)
    75.75
    >sd(score)
    11.92465
(d) > quantile(score, c(0.5, 0.25, 0.75))
    50%  25%  75%
    75.50 67.75 85.50
(e)
    > quantile(score, 0.8)
    80%
    87.2
(f) > IQR(score)
    17.75
(g) > boxplot(score)

```

